SIVASAI CHARY MUTYALA

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PROFESSIONAL SUMMARY

System Assurance Engineer / RAMS Engineer with 2 years of experience in the railway sector, supporting RAMS activities for Rolling Stock, Signalling, Telecom, Power Supply & OHE. Proficient in RAM analysis and prediction, FMEA/FMECA, RAM demonstration, hazard and risk assessment and Life Cycle Cost (LCC) analysis in accordance with CENELEC standards (EN 50126). Experienced in developing RAMS deliverables, managing hazard logs, and working with multidisciplinary teams throughout the railway system lifecycle.

SKILLS

- RAM Analysis, Prediction & Demonstration
- Reliability Engineering (RBD, FMEA/FMECA)
- Maintainability, Spare Parts & Life Cycle Cost (LCC) Analysis
- Weibull Analysis
- PHA, SHA, IHA, Risk Assessment & Hazard Log Management
- Railway System Assurance & Lifecycle Management (SA & GA)
- CENELEC Standards (EN 50126)
- Supplier & Stakeholder Management
- Technical Documentation & Reporting
- Familiarity with DOORS
- Microsoft Office - Excel, Word & PowerPoint

EXPERIENCE

ASCRIT Private Limited – *System Assurance Engineer* ***OCT 2024 - Present***

- Currently working as a RAMS Engineer for the Vande Bharat Train project, supporting RAM Apportionment and Prediction, Hazard Analysis (PHA, SHA), Hazard Log management, and supplier/subsystem coordination activities to meet system-level RAMS requirements.
- Working as a RAM Engineer for the DMRC Phase-4 (Telecom), TRAC-GAPSD (Generic Application PSD), and MML5 Signalling projects, overseeing RAM activities and supporting the preparation and review of RAM deliverables, including RAM Plan, FMECA, RAM Analysis & Prediction, and RAM Demonstration Plans and Reports.
- Worked as a RAMS Engineer for the GMRC Phase-1 (Telecom) project, performing RAM activities throughout the system lifecycle and supporting the Safety deliverables. Also contributed to Safety Assurance activities for Stages 3 and 4 of the RRTS (PST & OHE) project.
- Supported Interface Hazard Analysis (IHA) for the Rolling Stock–Signalling interface of the GMRC Phase-2 (Signalling) project by identifying interface hazards, assessing associated risks, and supporting the implementation of mitigation measures. Also performed RAM Analysis & Prediction for the Signalling system.
- Performed RAM Analysis & Prediction and Reliability Block Diagram (RBD) modelling to assess system performance and verify compliance with contractual RAM requirements.
- Developed and maintained FMEA/FMECA, Performance Monitoring, RAM Demonstration, and other RAMS deliverables in accordance with project requirements and applicable standards.

- Performed performance monitoring and RAM demonstration activities for the GMRC Phase-1 (Telecom) project through FRACAS-based analysis of operational and failure data to validate reliability, availability, and maintainability targets.
- Performed PHA, SHA, IHA, and risk assessment activities to identify, evaluate, and mitigate system and interface-related hazards.
- Performed Maintainability Analysis (MA, SMA, MTA & LORA) and Spare Parts Analysis to optimize maintenance planning and support system availability objectives.
- Performed Obsolescence Management analysis for Vande Bharat Train subsystems, including Air Duct, Interior Lighting, Pantry, and Sanitary systems.
- Conducted Life Cycle Cost (LCC) Analysis, including Preventive Maintenance (PM), Corrective Maintenance (CM), and spare parts cost assessments to support cost-effective asset management decisions.
- Performed Weibull Analysis to evaluate reliability characteristics and predict failure behaviour of Vande Bharat Train interior lighting components, supporting reliability improvement and maintenance optimization initiatives.
- Collaborated with multidisciplinary teams, project engineers, suppliers, OEMs, and key stakeholders to support RAMS and Safety Assurance activities and ensure successful project delivery.

EDUCATION

Bachelor of Technology (Electrical & Electronics Engineering) *2021 - 2024*
RVR & JC College of Engineering **CGPA: 7.8/10**

Diploma (Electrical & Electronics Engineering) *2018 – 2021*
AANM & VVRSR Polytechnic **Percentage – 93.4%**

Class 10th – SSC *2017 – 2018*
St. Johns E.M. High School **CGPA: 9.2/10**